

MoMoneyMoProblems Sprint 1

FR#	Short title	Description	Priority	In scope this semester?	Source
FR1	Add expense	The system should allow users different expenses they pay from month to month	high	Yes	Team assumption
FR2	Add income stream	The system should allow users to add different ways they make money	high	yes	Team assumption
FR3	Edit income stream	The system should allow it so users can update the amount of money they are making	high	yes	Team assumption
FR4	Edit expense	The system should allow it so users can update the amount they want to spend in a month	high	yes	Team assumption
FR5	Delete income stream	The system should allow it so users can remove an income stream they already added	high	yes	Team assumption
FR6	Delete expense	The system should allow	high	yes	Team assumption

		it so users can remove an expense they already added			
FR7	Set budget	The system should allow users to set a goal of how much money they want to spend a month	high	yes	Team assumption
FR8	Recommend budget	Based on expenses and income stream, the system should recommend a categorized budget for the month	low	no	Customer interview
FR9	View spending	The system should track your spending each month	medium	no	Team assumption
FR10	Create savings goal	The system shall make it so a user can track how much money they want to save out of their income every month	medium	yes	Team assumption
FR11	Edit savings goal	The system shall make it so a user can edit their savings goal	medium	yes	Team assumption

		month to month			
FR12	delete savings goal	The system shall make it so a user can delete savings goals month to month	medium	yes	Team assumption

Measured	Target	Condition
The load time of any page	Under 1 second	When a user opens the website
The accuracy of the budget calculations	100%	When a user calculates their budget for the month

Constraint

C1, Banking Services, Because not all banking services have API the development team can access, we cannot automatically update the budgeting app with expenses and income streams.

Scenarios

John is a college student who makes \$250 every two weeks working at a pizza shop, and an additional \$500 a month doing freelance coding. John pays \$400 per month in rent and spends about \$120 per week on groceries. At the start of each month John checks a banking app to see how much money is available but he does not have a specific monthly budget. Throughout the month John buys groceries, pays rent, eats at restaurants, and spends money on entertainment. John occasionally checks his bank account balance but does not keep track of how much has been spent. Near the end of the month John realizes that more money has been spent than expected. John looks through past transactions and tries to determine where the money went. Because the expenses and income were not organized into a budget, John has difficulty determining how much money can be spent while leaving enough for upcoming expenses.

John is a college student who makes \$250 every two weeks working at a pizza shop, and an additional \$500 a month doing freelance coding. John pays \$400 per month in rent and spends about \$120 per week on groceries. At the start of the month John uses MoMoneyMoProblems to record both sources of income and the expected expenses of the month. John creates a monthly budget and a savings goal. As expenses occur throughout the month, John adds them to MoMoneyMoProblems to keep track of what he is spending. John is able to add his money from the pizza shop and freelance coding to keep track of how much he is making. As he is going out to shop for groceries, eat at restaurants and spend money on entertainment, John can keep track of everything he spends and now knows how much he can spend at each place. At the end of the month, John can look at everything he spent and made and use that to plan for the next month.

Use case:

Use case name: Add Expense

Participating Actors: User

Flow of Events:

1. The user clicks on button "Add Expense"
 - a. The system takes you to a new page and requests the amount of money and category for expense
2. User inputs the amount of money spent and category
 - a. The system validates the user inputs
3. User clicks "confirm add expense"
 - a. The system adds expenses to other expenses and recalculates needed numbers.

Entry Condition: The user has opened the app

Exit Condition: The user successful adds the expense to their budget

Quality Requirements: The app successfully calculates the new budget with 100% accuracy.

Traceability

Requirements	Scenario	Use Case	Comments
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FR1 - Add Expense	Visionary Scenario	UC1(Add Expense)	John records expected expenses at the start of the month and adds new ones as they happen
FR2 - Add Income Streams	Visionary Scenario	-	John record pizza shop and freelance income. UC only covers expenses
FR3 - Edit Income Streams	-	-	Not covered, since John doesn't change his income amount
FR4 - Edit Expense	-	-	Not covered, no scenario where john makes changes an entry he already made
FR5 - Delete Income Streams	-	-	Not covered, no scenario where john stops working
FR6 - Delete Expense	-	-	Not covered, no scenario where john removes an expense he created
FR7 - Set Budget	Visionary Scenario	UC1(Add Expense)	John creates a monthly budget. The UC calculates the budget after expenses
FR8 - Recommend Budget	-	-	Not covered, since this may be out of scope for the semester
FR9 - View Spending	Visionary Scenario	-	The the end of the month John reviews what he spent and earned
FR10 - Create Saving Goals	Visionary Scenario	-	John sets a savings goal in the beginning of the month
FR11 - Edit Saving	-	-	Not covered, John

Goals			doesn't change his goals month to month
FR12 - Delete Saving Goals	-	-	Not covered, John doesn't change his goals month to month
NFR1 - Loading page in under 1 second	-	UC1(Add Expense)	Applies when the system opens a page. Scenarios don't address performance
NFR2 - 100% calculation accuracy	Visionary Scenario	UC1(Add Expense)	Matches the UC's quality requirements. In the scenario John relies on accurate totals to know how much he can spend

Customer Elicitation

We interviewed Krish Batra who is a Computer Science student at the University of Pittsburgh who manages his own spending on food, groceries, transportation, entertainment, and school supplies. The interview was conducted by Nathan, Graham, and Momrez on September 14th in person and lasted about 15 minutes.

Questions/Answers:

- How do you currently keep track of your spending, and what is the most frustrating part of it?
 - **I have a spreadsheet that I manually have to edit every week and cross check with the card expenses**
- Do you want one overall monthly budget, or separate budgets for categories like food, transportation, and entertainment?
 - **I would really like to have separate budgets for different categories and and for overall**
- When you set a budget, would you rather choose the amounts yourself or have the app suggest them based on your income and expenses?
 - **Having the app set it for me**
- When you log an expense, what details matter to you: just the amount and category, or also the date and where you spent it?
 - **The date and where I spend my money will give me a good understanding of how I spend the money so I can budget properly**
- Is it important to look back at what you spent in past months, or do you only care about the current month?
 - **I don't care too much about previous spending but it's good to look at my habits so I can be more responsible in the future, so if something can tell**

me that I usually spend over my food budget every month then I would dial it back on eating out as much.

Requirements Changed:

Requirements	Before	After
FR8 - Recommend Budget	Not Included	The system should recommend a categorized monthly budget based on the user's expenses and income streams.
FR4 - Edit Expense	Not Included	The system show allow to edit the budget depending on end user preference

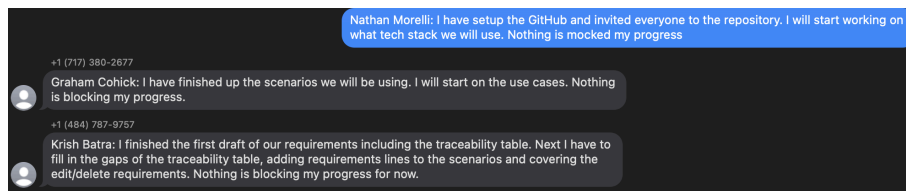
Something that surprised us:

- Something that surprised us was that Krish said he would like to see his previous month's spending so he could see how he spends his money and he could work his budget around his habits.

Validation Conversation:

- Since our customer is a team member, we also spoke with Arron, another Pitt student who pays their own rent and groceries, on October 1st. We asked him to validate whether our in-scope requirements match how they handle money and whether manually entering expenses is realistic. He confirmed all the requirements would be beneficial for him.

Scrum Process Evidence





Krish talked about the git workflow that he was researching so we don't have conflicts when we push to the repository. Nathan was talking about the framework we decided on and how he was starting to set up the repository for that. Graham talked about the vision of the app and what he thought was most important for us to get done this week. We had no conflicts.

Reflection:

Krish Batra Dev

I reviewed the finalized requirements and the use case to estimate the inscope requirements features and began to set up the project's basic structure starting with code functionality.

Nathan Morelli Dev

I set up the github and also chose the technology to use for this application. For this application, we chose a website for ease of use so we are using the Next.js framework.

Momrez Mohsin PO

I revised and reprioritized the product backlog based on the customer interview deciding which requirements change, whether FR9 moves into scope, and which features the team should build first.

Graham Cohick SM

I set up the times for the scrum meetings. I also drafted out the scenarios and use case based on our initial requirements.